

Maziar Zamanpour

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SUMMARY

Ph.D. candidate (graduating Jan 2027) with 4+ years of experience in supply chain analytics, optimization, and predictive modeling using Python, SQL, and Power BI. Industry experience analyzing ERP/MRP data, root-causing on-time delivery misses, and building planning and forecasting tools adopted by operations leaders. Skilled at translating complex data into strategic, actionable recommendations.

TECHNICAL SKILLS

- **Optimization:** Mixed-Integer Programming (Gurobi, CPLEX, PuLP), Model Predictive Control, Dynamic Programming, Gradient Descent, Greedy & Heuristic Algorithms.
- **Programming:** Python (NumPy, Pandas, scikit-learn, Gurobipy), SQL (T-SQL), Java, R.
- **Analytics & ML:** Forecasting (ETS, gradient boosting), Regression, Predictive Modeling, Anomaly Detection, Databricks, Power BI (DAX), Excel, PowerPoint.
- **Tools:** ERP/MRP (SAP), SQL Server, Git/GitHub, SUMO, VISSIM, Microsoft Office, Large Language Models (AI).

EXPERIENCE

Graduate Data Analyst Intern — Operations & Supply Chain

Bio-Techne - Minneapolis, MN

May 2026 - present

- Identified key drivers of planning and manufacturing delays and **automated root-cause attribution** of on-time delivery (OTD) misses in data pipelines (SQL, Power BI), replacing manual review across six delay categories (production, QC, planning, lead time, shipping).
- Built operational decision tools — **historical stock lookup, demand profiles, and lead-time benchmarks** — from ERP/MRP data, used directly by planners and supply chain directors; contributed to a **major OTD improvement**.
- Uncovered and resolved systemic data-quality gaps (e.g., integrity issues), improving reliability of planning and inventory analytics.
- Developed **machine-learning forecasting models** (Python) for demand and OTD-miss risk, flagging at-risk open orders before promise dates and informing **stock policy** (Databricks).

Operations Research & Optimization — Research Scientist

University of Minnesota - Minneapolis, MN

August 2022 - present

- Developed **real-time model-predictive control** system for vehicle red light running violation based on vehicle connectivity; deployed on public roads in Minnesota with **safety enhancement proved** (GitHub).
- Calibrated driver behavior models using **gradient descent** and short-term prediction, and built a **supervised ML pipeline** predicting lane-change behavior from trajectory data; delivered individualized recommendations validated on real-world data.
- Conducted **cost-benefit analysis** for MN DOT; demonstrated payback within 3 years, informing deployment decisions.

Optimization & Product Development — Entrepreneurial Lead

National Science Foundation I-Corps Program

May 2024 - present

- Built dispatch optimization (**integer programming, CPLEX**) and ML demand forecasting for a car-sharing venture; proved **75% cost reduction** vs. personal vehicle ownership, with operational modules configured in **SAP**.
- Secured **\$50K grant**; led 100+ customer discovery interviews to validate product-market fit.

Statistics and Probability — Course Instructor

University of Minnesota - Minneapolis, MN

May 2024 - August 2025

- Taught probability, causal inference, Bayesian methods, distributions, point estimation, regression, and forecasting.

EDUCATION

University of Minnesota - Minneapolis, MN

Dec 2023 – Jan 2027

Ph.D., Civil Engineering: Transportation (+ Industrial Engineering Minor)

GPA 3.7

Coursework: Optimization for ML, Optimization for Data Science, Production & Inventory Systems, Decision Analysis, Storytelling for Engineers

University of Minnesota - Minneapolis, MN

Aug 2022 – Dec 2023

M.S., Civil Engineering: Transportation

GPA 3.6

University of Tehran - Tehran

Sep 2018 – May 2022

B.S., Civil Engineering

GPA 3.7

SELECTED PROJECTS

DoorDash Delivery Optimization

Fall 2025

- Formulated driver-delivery assignment as a **MIP** (Gurobi); designed greedy heuristic for real-time dynamic assignment.

PUBLICATIONS & AWARDS

- **7 publications** (3 peer-reviewed journals incl. Transportation Research Part C; 2 state DOT reports; 2 under review). Presented at INFORMS, TRB, ITS Minnesota.
- **Matthew J. Huber Award** for excellence in transportation research (2024).
- Won two **TRB Travel Awards** (2023, 2025).